ACTL2102 Tutorial

Week 7: Compound Poisson Processes and Continuous-Time Markov Chains

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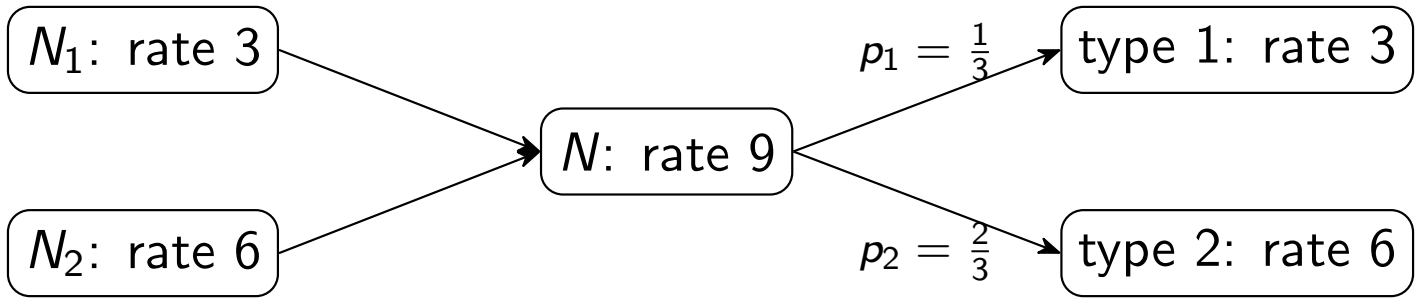
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Plan

- Recap: superposition and thinning, the compound Poisson process, continuous-time Markov chains (holding times and jumps, transition rates, the generator, the Kolmogorov equations)
- Main questions: **3.13** and **3.14** (parts 1–2, revisited briefly), **3.16, 4.2, 4.11**
- If time permits: 4.1

Superposition and Thinning



superposition: rates add thinning: the rate splits

Superposition. Independent Poisson processes with rates λ_1 and λ_2 add to a Poisson process with rate $\lambda_1 + \lambda_2$. Each event of the sum comes from the first process with probability $\lambda_1/(\lambda_1 + \lambda_2)$, independently of the event times and of the other events.

Thinning. If each event of a Poisson process with rate λ is, independently of all others, of type k with probability p_k, for mutually exclusive types $k = 1, \ldots, m$, then the type-k events form independent Poisson processes with rates λp_k.

Exercise 3.13: Two claim types: which type was this claim?

Exercise 3.13: [poi9, Solution] Ross: Chapter 5, Q59. There are two types of claims that are made to an insurance company. Let $N_i(t)$ denote the number of type i claims made by time t, and suppose that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes with rates $\lambda_1 = 10$ and $\lambda_2 = 1$. The amounts of successive type 1 claims are independent exponential random variables with mean \$1000 whereas the amounts from type 2 claims are independent exponential random variables with mean \$5000. A claim for \$4000 has just been received; what is the probability it is a type 1 claim.

Exercise 3.14: Portfolio takeover: the claim processes (parts 1–2)

Exercise 3.14: [poi10, Solution] Insurer A has a combined home insurance and landlords insurance portfolio. The total number of claims for this portfolio is modelled using a Poisson process with expected claims 300 per year. The proportion of landlords insurance claims was one fifth of the overall claims. Insurer A sells its home insurance portfolio to insurer B. Insurer B specializes in home insurance and has no landlords insurance policies. The expected number of claims for the old portfolio of insurer B was 120 and can also be modelled using a Poisson process. The number of claims for insurer A is independent of the number of claims for insurer B.

1. Define the processes of the number of claims from insurer A for both the home insurance portfolio and landlords insurance portfolio before the takeover.
2. Define the processes of the number of claims of both insurer A and B before and after the takeover of the home insurance by insurer B.

The Compound Poisson Process

Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ, and let $Y_1, Y_2, \ldots$ be i.i.d. random variables, independent of $\{N(t)\}$. The **compound Poisson process** is $X(t) = \sum_{i=1}^{N(t)} Y_i$, $t \geq 0$, with $X(t) = 0$ when $N(t) = 0$.

Given $N(t)$, the mean of $X(t)$ is $N(t)\,\mathrm{E}[Y_1]$ and the variance is $N(t)\,\mathrm{Var}(Y_1)$. Conditioning on $N(t)$, with $\mathrm{E}[N(t)] = \mathrm{Var}[N(t)] = \lambda t$,

$$\mathrm{E}[X(t)] = \lambda t\ \mathrm{E}[Y_1],$$
$$\mathrm{Var}[X(t)] = \lambda t\,\mathrm{Var}(Y_1) + \lambda t\,(\mathrm{E}[Y_1])^2 = \lambda t\ \mathrm{E}[Y_1^2].$$

Example. Claims arrive at 22 per month and are for \$10k, \$5k or \$1k with probabilities 0.05, 0.15 and 0.80. Then $\mathrm{E}[Y_1^2] = 9.55$ and the monthly total has variance $22 \times 9.55 = 210.1$, both in $(\$k)^2$.

Exercise 3.16: Life insurance claims: mean and variance of four weeks' payments

Exercise 3.16: [poi12, Solution] Ross: Chapter 5, Q85. An insurance company pays out claims on its life insurance policies in accordance with a Poisson process having rate $\lambda = 5$ per week. If the amount of money paid on each policy is exponentially distributed with mean \$2000, what is the mean and variance of the amount of money paid by the insurance company in a four-week span?

Continuous-Time Markov Chains

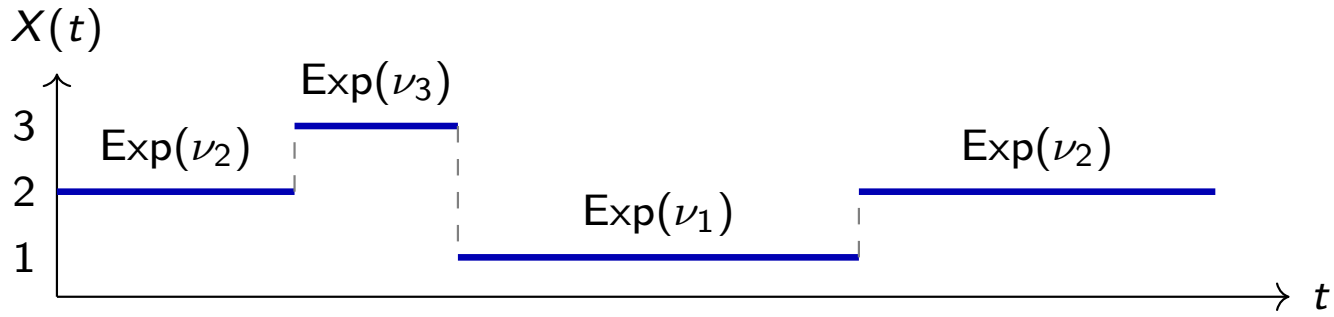
A **continuous-time Markov chain** $\{X(t), t \geq 0\}$ on a discrete state space S can change state at any time, and has the Markov property: for all $s, t \geq 0$, all states i, j and every history $\{x(u), 0 \leq u < s\}$,

$$\begin{aligned}\Pr[X(s+t) = j \mid X(s) = i,\ X(u) = x(u) \text{ for } 0 \leq u < s] \\ = \Pr[X(s+t) = j \mid X(s) = i].\end{aligned}$$

The chain is **homogeneous** if this probability depends only on t. It is then the **transition probability** $P_{ij}(t)$: the probability of being in j a time t after being in i, whatever happens in between.

A discrete-time chain is described by one matrix $\mathbf{P}$. Here there is a matrix $P(t) = [P_{ij}(t)]$ for every $t \geq 0$, rarely available in closed form. A continuous-time chain is therefore specified by its rates of moving between states, collected in the generator Q, and $P(t)$ is obtained from Q.

Holding Times and Jumps



Each visit to state i lasts a **holding time** T_i. By the Markov property T_i is memoryless, since the time already spent in i says nothing about the time remaining. So $T_i \sim \text{Exp}(\nu_i)$, where ν_i is the rate of leaving i and $\text{E}[T_i] = 1/\nu_i$.

Given that the chain leaves i, it moves to $j \neq i$ with probability

$$P_{ij} = \Pr(\text{next state is } j \mid \text{the chain leaves } i), \qquad \sum_{j \neq i} P_{ij} = 1,$$

independently of T_i. This concerns the next state only; $P_{ij}(t)$ is the probability of being in j at time t, after any number of jumps.

Transition Rates

The chain leaves i at rate ν_i, and a proportion P_{ij} of its departures from i go to j. The **transition rate** from i to j is the rate of moves from i to j:

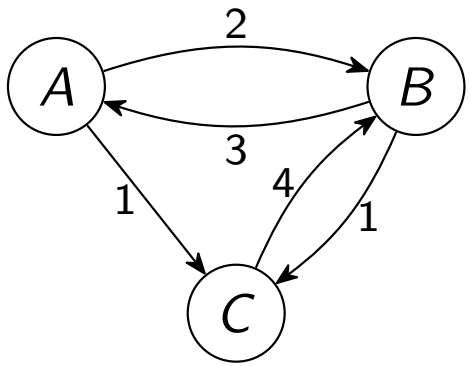
$$q_{ij} = \nu_i P_{ij}, \qquad j \neq i.$$

Summing over j gives $\nu_i = \sum_{j \neq i} q_{ij}$, and dividing gives $P_{ij} = q_{ij}/\nu_i$. The rates therefore carry the same information as the holding times and the jumps.

	$P_{ij}(t)$	P_{ij}	q_{ij}
meaning	in j at time t, from i	next state j, given a jump from i	rate of moves $i \to j$
values	in $[0,1]$	in $[0,1]$, with $P_{ii} = 0$	any value ≥ 0
row sums	1	1	0, with $q_{ii} = -\nu_i$

The Generator

The **generator** $Q = [q_{ij}]$ collects the rates, with $q_{ii} = -\nu_i$ on the diagonal so that each row sums to zero. Together with the initial state, Q specifies a homogeneous chain. It is usually drawn as a diagram with the rates on the arrows.



$$Q = \begin{pmatrix} -3 & 2 & 1 \\ 3 & -4 & 1 \\ 0 & 4 & -4 \end{pmatrix}$$

rows and columns in the order A, B, C

So Q determines the holding times ($\nu_i = -q_{ii}$), the jumps ($P_{ij} = q_{ij}/\nu_i$) and, through the Kolmogorov equations, $P(t)$. Here a visit to A lasts $1/3$ on average and ends in B with probability $2/3$. Constant rates give a homogeneous chain; rates $q_{ij}(t)$ that change with time give a non-homogeneous chain (Week 9).

Rates over a Short Time

The chain leaves i within a short time h if $T_i \le h$. By the Taylor series $e^{-x} = 1 - x + \frac{1}{2}x^2 - \cdots$, this has probability

$$\Pr(T_i \le h) = 1 - e^{-\nu_i h} = \nu_i h - \tfrac{1}{2}(\nu_i h)^2 + \cdots = \nu_i h + o(h),$$

where $o(h)$ stands for terms with $o(h)/h \to 0$ as $h \to 0$, negligible next to h. For example, with $\nu_i = 2$ and $h = 0.01$,
$1 - e^{-0.02} = 0.0198$, against $\nu_i h = 0.02$.
Two or more jumps need two holding times shorter than h, which has probability $o(h)$, and a single jump goes to j with probability P_{ij}. So

$$\begin{aligned} P_{ij}(h) &= \nu_i h\, P_{ij} + o(h) = q_{ij} h + o(h), \qquad j \ne i, \\ P_{ii}(h) &= 1 - \nu_i h + o(h). \end{aligned}$$

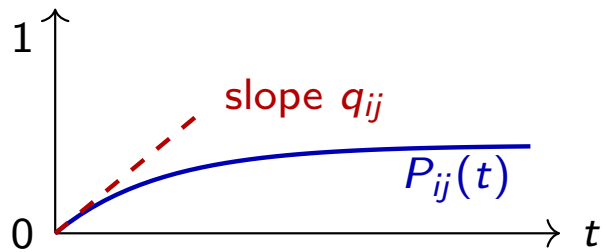
A move from i to j in a short time h therefore has probability about $q_{ij} h$.

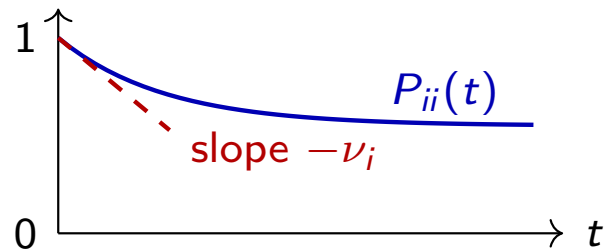
Rates as Slopes at Time 0

At time 0 the chain is still in its starting state, so $P(0) = I$. Dividing the expansions by h,

$$\frac{P_{ij}(h) - P_{ij}(0)}{h} = q_{ij} + \frac{o(h)}{h} \;\to\; q_{ij},$$
$$\frac{P_{ii}(h) - P_{ii}(0)}{h} = -\nu_i + \frac{o(h)}{h} \;\to\; -\nu_i = q_{ii}$$

as $h \to 0$. These limits are derivatives at $t = 0$: each rate is the slope with which the corresponding transition probability starts.



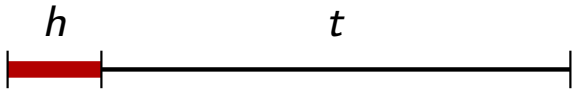


In matrix form, $P(h) = I + Qh + o(h)$, so Q is the slope of $P(t)$ at $t = 0$. The lecture introduces q_{ij} through this slope, which agrees with $q_{ij} = \nu_i P_{ij}$.

The Kolmogorov Equations

A path from i to j over time $t+s$ is in some state k at time t, so $P(t+s) = P(t)P(s)$, the **Chapman–Kolmogorov equations**.

Backward equations



$$P(t+h) = P(h)\,P(t)$$
$$= P(t) + hQP(t) + o(h)$$

$$\frac{\partial}{\partial t}P(t) = Q\,P(t),$$

$$\frac{\partial}{\partial t}P_{ij}(t) = \sum_k q_{ik}\,P_{kj}(t).$$

Forward equations



$$P(t+h) = P(t)\,P(h)$$
$$= P(t) + hP(t)Q + o(h)$$

$$\frac{\partial}{\partial t}P(t) = P(t)\,Q,$$

$$\frac{\partial}{\partial t}P_{ij}(t) = \sum_k P_{ik}(t)\,q_{kj}.$$

Subtracting $P(t)$, dividing by h and letting $h \to 0$ gives each equation, starting from $P(0) = I$, with Q on the side of the short piece. The Poisson process with rate λ is the chain on $\{0, 1, 2, \dots\}$ with only the transitions $i \to i+1$, each at rate λ.

Exercise 4.2: Poisson process: the Kolmogorov forward equations

Exercise 4.2: [cmc1, Solution] The total number of claims received by an insurance company is modelled according to a time-homogeneous Poisson process with rate λ.

1. Write down the Kolmogorov forward equations for this time-homogeneous process.
2. Show that the solution is of the form:

$$P_{0j}(t) = \frac{(\lambda t)^j \cdot e^{-\lambda t}}{j!}.$$

Exercise 4.11: Two machines and one repairman: the backward equations

Exercise 4.11: [cmc10, Solution] Ross: Chapter 6, Q8. Consider two machines, both of which have an exponential lifetime with mean $1/\lambda$. There is a single repairman that can service machines at an exponential rate μ. Set up the Kolmogorov backward equations; you need not solve them.

Exercise 4.1: Force of mortality and the distribution of the age at death

Exercise 4.1: [cmcpre1, Solution] (ACTL2131) Actuaries often model the age-at-death as a non-negative random variable X and define the 'force of mortality' as follows:

$$\mu(x) = \lim_{h \to 0} \frac{F_X(x+h) - F_X(x)}{h(1 - F_X(x))} \quad ,$$

where $F_X(\cdot)$ denotes the cdf of X. Using this definition, prove that:

$$F_X(x) = 1 - \exp\left(-\int_0^x \mu(z)\, dz\right).$$

Thanks for coming!